

Privacy-Preserving Integrity Proofs for Student-Society Voting: A Phase C Pilot of Project Simurgh at Macquarie University

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Abstract—We show that a consent-based shadow-mode system can collect session-integrity metadata alongside a real voting event without recording ballot choices — enforced structurally, not by policy alone. Digital voting tools raise an integrity question that existing cryptographic protocols (end-to-end verifiable voting, mix-nets) do not address: can a participant verify their session was conducted fairly without revealing their vote? Project Simurgh answers this with an HMAC audit chain, client-side ballot-field discard before each network call, and a server-side forbidden-field guard that rejects any ballot-choice key. A 30-session Phase C pilot (31 consented, 1 withdrawn) confirmed `ballot_choice_recorded_by_simurgh: false` in every submitted session; 359/359 automated tests, 8/8 smoke gates, and 10/10 security-audit gates passed at closeout. After data collection ended, all write endpoints returned HTTP 410 Gone under a server-side collection-closed flag, verifiable by any caller. This study is voting-adjacent: it does not implement ballot cryptography, voter eligibility verification, coercion resistance, tally protection, or public-election certification.

Index Terms—privacy-preserving integrity, shadow-mode pilot, student-society voting, HMAC audit chain, consent-based telemetry, collection closure

I. INTRODUCTION

Student-society governance increasingly relies on digital voting tools for event preference polls, committee elections, and budget decisions. These tools introduce a question existing cryptographic solutions leave open: can a participant confirm their session was handled honestly — no screen capture, no ballot recording, no passive surveillance — without revealing their choice?

End-to-end verifiable voting schemes [1], [2] solve the *outcome* integrity problem (did the tally reflect all ballots?) but say nothing about *session* integrity (was the participant’s interaction free of covert collection?). Project Simurgh is a research prototype that targets session integrity through privacy-preserving metadata collection and tamper-evident audit chains [3].

Voting-adjacent case study, not a production election-security claim. Research prototype only. Source: <https://github.com/Raouf128/Project-Simurgh>. Tag: `v0.5.0-voting-pilot-phase-c-closeout`.

This paper reports a Phase C pilot in which Project Simurgh ran in shadow mode alongside a real MQ Persian Society event preference poll: collecting session-level integrity metadata in parallel, with no effect on the official result. The pilot was not a study of election security, voting-system integrity, or electoral fraud prevention. The official election result was determined entirely by the society’s own voting system.

Contributions:

- A consent-to-submit pilot with 30 submitted sessions confirming `ballot_choice_recorded_by_simurgh: false` in every session, backed by smoke, security-audit, and privacy-audit gate suites (§V).
- A server-side collection-closure mechanism returning HTTP 410 Gone on all write routes, enforceable and verifiable independently of the client UI (§III).
- A reproducible evidence baseline at tag `v0.5.0-voting-pilot-phase-c-closeout` for future work on privacy-preserving integrity proofs in voting-adjacent contexts.

II. RELATED WORK

A. End-to-End Verifiable Voting

Systems such as Helios [1] use homomorphic encryption or mix-nets so voters can verify their encrypted ballot was counted without revealing the plaintext choice. Civitas [2] adds coercion resistance; STAR-Vote [4] adds risk-limiting-audit integration.

Project Simurgh differs in scope. It does not implement ballot encryption, voter-verifiable tallying, or public election certification. Where end-to-end verifiable voting proves the integrity of the *outcome*, Project Simurgh studies the integrity of the *participation session* — whether the system collected only what it disclosed and nothing more.

B. Remote and Internet Voting Security

Remote electronic voting introduces risks beyond ordinary web application security: endpoint compromise, coercion, eligibility verification, ballot secrecy, and tally integrity. NIST

guidance [5] identifies threat categories for remote electronic voting and recommends caution around internet-connected election infrastructure. The National Academies report [6] emphasizes that vote collection, alteration, destruction, counting, and reporting each require separate mitigations.

This pilot addresses none of those concerns. It is a student-society research case study, not a production online-voting system. It does not address eligibility, coercion resistance, malware-compromised endpoints, official tally integrity, or public-election certification.

C. Voting Standards, Privacy, and Auditability

The U.S. Election Assistance Commission adopted VVSG 2.0 in 2021 [7], establishing updated requirements for privacy, accessibility, auditability, and software independence in voting systems. Modern voting-system guidance consistently requires audit records that do not reveal individual ballot choices.

Project Simurgh aligns with this direction — the HMAC audit chain records session lifecycle events without ballot content — but remains outside public-election certification scope.

D. Australian Technology-Assisted Voting

Technology-assisted voting has been explored in several Australian jurisdictions, including iVote in New South Wales. The NSW Electoral Commission’s technology-assisted voting review [8] examined privacy, accessibility, and security trade-offs, recommending careful scoping and ongoing evaluation before broader deployment.

Australia’s cautious posture toward technology-assisted voting makes small, bounded pilots valuable for studying consent, privacy boundaries, and session-integrity evidence without affecting official outcomes.

E. Privacy-Preserving Telemetry and Data Minimisation

Project Simurgh’s ballot-field exclusion architecture follows data-minimisation principles [9], [10]: the system collects the minimum metadata required for a session-integrity signal and structurally prevents ballot-choice data from reaching the server. Data that is never collected cannot be leaked.

F. Position of This Work

This paper contributes a bounded empirical evaluation of privacy-preserving integrity evidence around a voting-adjacent workflow. The system remains blind to ballot choice and holds a non-disciplinary posture. The contribution is not a new voting protocol; it is evidence that shadow-mode session-integrity collection is achievable at student-governance scale.

III. SYSTEM DESIGN

Fig. 1 shows the Phase C session lifecycle.

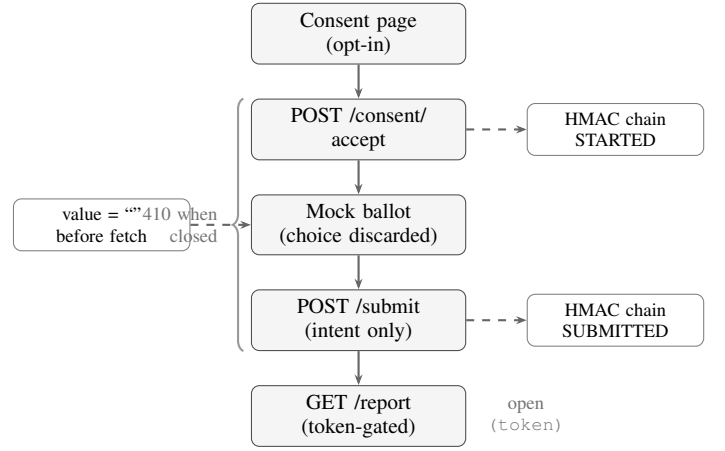


Fig. 1. Phase C session lifecycle. The ballot choice is discarded in the browser before every `fetch` call (left). HMAC audit-chain events are appended on consent and submit (right). Under the collection-closed configuration, consent/accept, submit, and withdraw return HTTP 410 (bracket); the report endpoint remains active and token-protected.

A. Consent Flow

The consent page disclosed the data categories Project Simurgh may collect across its deployments — session timestamps, focus-loss and paste counts, audit-chain validity signal, and an anonymous HMAC-hashed session code — and what it never collects: ballot choices, screen recordings, webcam/audio, typed or pasted content, raw process names, window titles, or device serial identifiers. In Phase C, implemented collection comprised session timestamps, audit-chain lifecycle events, and an anonymous participant code; focus-loss and paste counts are exam-integrity features of the broader Project Simurgh system and were not active in the voting pilot. Participation was voluntary; declining produced no consequence.

On consent, the server issued an anonymous participant code and a signed session token. No personally identifying information was collected at any step.

B. Ballot-Field Exclusion

Before every `fetch` call the submit page set all radio button value attributes to an empty string, ensuring the ballot field was structurally absent from the request body. The server provided a second layer: the `FORBIDDEN_BALLOT_FIELDS` set rejected any submission body containing a known ballot-choice key with HTTP 400, with no echo of the supplied value.

C. HMAC Audit Chain

Each session maintained an append-only HMAC audit chain recording lifecycle events: `STARTED`, `SUBMITTED`, and optionally `WITHDRAWN` or `REPORT_EXPORTED`. The chain initialized on consent and verified on report export.

D. Collection-Closure Posture

After the target dataset was reached, we set `SIMURGH_VOTING_PILOT_COLLECTION_CLOSED=true`.

With this flag active:

- `POST /api/voting-pilot/consent/accept` returns 410 Gone.
- `POST /api/voting-pilot/submit` returns 410 Gone.
- `POST /api/voting-pilot/withdraw` returns 410 Gone.
- `GET /api/voting-pilot/:id/report` remains available, token-protected.

The closure middleware fires before authentication on write routes; a valid session token cannot bypass it.

IV. PHASE C PILOT

A. Study Design

The pilot ran in shadow mode alongside the MQ Persian Society 2026 event preference poll — an event asking members to rank four proposed social events (Nowruz cultural night, Persian movie night, career/networking night, food and music night). Project Simurgh ran in parallel without access to, or influence on, the official ballot-counting system. Each participant who chose to join the pilot completed two steps: a consent page explaining data collection and an optional mock ballot page that mirrored the official event poll. The mock ballot choice was discarded in the browser before any network call; Project Simurgh received only a submit-intent signal.

Participants were members of the society. Participation was voluntary and governed by the Phase C participant notice and data management addendum approved by the society executive on 2026-06-04. No incentive was offered.

B. Session Counts

Of 31 sessions that accepted the consent flow, 30 completed the submit step and form the primary analysis dataset. One session (`vp_4fcc741a`) was withdrawn by the participant before submit and is excluded from all analysis; its report endpoint returns HTTP 403.

C. Data Management

The server held session data in memory only — no data persisted to disk, and the store cleared on server restart. This is appropriate for a research prototype and consistent with the data management addendum, which specified no persistent storage of pilot session records.

V. RESULTS

A. Dataset

B. Privacy Assertions

The `ballot_choice_recorded_by_simurgh` field is a behavioral guarantee enforced by both client-side discard and server-side forbidden-field rejection. Because the session store was in-memory and cleared after collection, individual session reports are not available post-hoc; the guarantee is verified by the automated smoke suite at every gate run.

TABLE I
SESSION COUNTS

Metric	Count
Consented pilot sessions	31
Submitted sessions (primary analysis)	30
Withdrawn sessions	1
Official vote impact	0
Ballot choices collected by Project Simurgh	0

TABLE II
PRIVACY ASSERTIONS — ALL 30 SUBMITTED SESSIONS

Assertion	Result
<code>ballot_choice_recorded_by_simurgh</code>	false
Ballot choice transmitted to server	No
Screen capture	No
Webcam / audio	No
Typed content	No
Pasted content	No
Personal device identifiers	No
Raw process names	No
Raw window titles	No
<code>official_vote_impact</code>	false
Privacy audit (tools/privacy-audit.mjs)	PASS (code + evidence files)

TABLE III
SAFETY GATES AT CLOSEOUT (TAG
V0.5.0-VOTING-PILOT-PHASE-C-CLOSEOUT)

Gate	Result
Node tests (npm test)	359/359 pass
npm audit (--audit-level=high)	0 high/critical
Privacy audit	PASS (52 evidence files)
<code>smoke-voting-pilot.sh</code>	8/8 pass
<code>security-audit-voting-pilot.sh</code>	10/10 pass
<code>smoke-voting-pilot-closed.sh</code>	5/5 pass

C. Integrity Flow

All 30 submitted sessions completed the consent-to-submit flow. The HMAC audit chain recorded `STARTED` on consent and `SUBMITTED` on submit for each. The withdrawn session's chain recorded `WITHDRAWN`; its report endpoint returned HTTP 403, confirming exclusion from analysis.

D. Safety Gates

Gate evidence files are archived at `docs/research/mq-voting-pilot/evidence/phase-c-clo` in the repository at the baseline tag.

E. Collection Closure

After the 30-session target was reached, all write endpoints returned HTTP 410 Gone. The closure smoke suite (5/5 gates) confirmed that `consent/accept`, `submit`, and `withdraw` are unreachable under the closed configuration, while `report` export remains available with a valid token.

VI. DISCUSSION

A. Interpretation

The pilot shows that a consent-based shadow-mode prototype can operate alongside a real student-society voting event

without collecting ballot choices or any other privacy-sensitive data. The privacy assertions in Table II hold by system design: the ballot field is structurally absent from the server request body in every session, not because participants complied with a policy.

The server-side collection-closure mechanism provides a reproducible, auditable end-of-collection posture. Any caller with network access can verify closure; the mechanism does not depend on the client UI.

B. Limitations

- The 30-session dataset is a proof-of-concept pilot. No statistical power calculation was performed; results should not be generalized beyond this scope.
- Sessions were anonymous; no demographic or behavioral analysis of participants is possible or intended.
- The pilot ran in shadow mode only. No claim extends to a non-shadow deployment.
- The in-memory store means session data did not survive server restarts. This suits a research prototype but is not a production data-management posture.
- `ballot_choice_recorded_by_simurgh: false` is a behavioral guarantee verified by the automated gate suite; per-session audit reports are not available post-hoc due to in-memory storage.

C. Non-Claims

This pilot does not claim to:

- Secure or audit the official MQ Persian Society election result.
- Detect or prevent election fraud.
- Provide hardware-rooted attestation.
- Replace or improve the society’s own voting system.
- Generalize to national, governmental, or binding elections.

VII. CONCLUSION

Thirty sessions completed the full consent-to-submit flow in Phase C of Project Simurgh. In every session, the system collected no ballot choice, screen recording, webcam or audio data, typed content, pasted content, or personal device identifier. After collection closed, all write endpoints returned HTTP 410 Gone under a server-side flag verifiable by any caller — proving that collection closure is not merely a UI gesture.

These results establish that privacy-preserving session-integrity evidence is structurally achievable alongside a real student-society voting event, without access to, or influence on, the official ballot.

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